1) 5x5 filter input is convolved with 3x3 filter, stride of 1 and no

$W_{in} = 5$

$H_{in} = 5$

$F = 3$

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$

$$= \frac{5 - 3 + 2 \times 0}{1} + 1$$

$$= \frac{2 + 0}{1} + 1$$

$$= 3$$

$$H_{out} = \frac{H_{in} - F + 2P}{S} + 1$$

$$= \frac{5 - 3 + 2 \times 0}{1} + 1$$

$$= \frac{2 + 0}{1} + 1$$

$$= 3$$

$\therefore$ Output = (3 x 3)

2) Compute the following

- Stride of 1, Filter = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Row 1 = C(1,1) = (20+35+35)+(29+44)+(16+25+21) = 225
C(1,2) = (35+35+35)+(46+42)+(25+21+19) = 258
C(1,3) = (35+35+35)+(44+42)+(21+19+19) = 260
C(1,4) = (35+35+20)+(42+27)+(19+19+12) = 209

Row 2 = C(2,1) = (29+46+44)+(16+21)+(66+120+116) = 458
C(2,2) = (46+44+42)+(25+19)+(120+116+154) = 566
C(2,3) = (44+42+42)+(21+19)+(116+154+114) = 552
C(2,4) = (42+42+27)+(19+12)+(154+114+62) = 472

Row 3 = C(3,1) = 708, C(3,2) = 981, C(3,3) = 887, C(3,4) = 802

Row 4 = C(4,1) = 1000, C(4,2) = 1488, C(4,3) = 1320, C(4,4) = 1224

$$\therefore C = \begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 981 & 887 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

3) Take the output from 1. and apply a max pooling layer with, Stride = [2,2], window shape = [2,2]

Now,

$$W_{out} = \left(\frac{win - F + 2P}{2}\right) + 1$$

$$= \left(\frac{4 - 2 + 2\times 0}{2}\right) + 1 = 2$$

Output shape = 2 x 2

Max pooling (Block 1) $\begin{bmatrix} 225 & 258 \\ 458 & 566 \end{bmatrix} = 566$

(Block 2) $\begin{bmatrix} 250 & 209 \\ 552 & 472 \end{bmatrix} = 552$, (Block 3) = 1488, (Block 4) = 1320

∴ Final output = $\begin{bmatrix} 566 & 552 \\ 1488 & 1320 \end{bmatrix}$

4) 2 x 2 filter, stride 2

$$W_{out} = \left(\frac{4 - 2 + 2\times 0}{2}\right) + 1 = 2$$

∴ Output shape = 2 x 2

Average pooling (Block 1) $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} = \frac{1+2+5+6}{4} = 3.5$,

(Block 2) $= \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} = 5.5$

(Block 3) $= \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix} = 11.5$

(Block 4) $= \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix} = 13.5$

∴ Final Output = $\begin{bmatrix} 3.5 & 5.5 \\ 11.5 & 13.5 \end{bmatrix}$

2. 5) To do

Filter = 3x3, Input = 5x5

- $W_{out} = \left(\frac{W_{in} - F + 2P}{2}\right) + 1 \left(\frac{5-3+2\times0}{2}\right) + 1 = 3 = 3\times3$

Ro $O_{12} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 2 & 2 \end{bmatrix} = \begin{matrix} 3\times0+2\times1+1\times2 \\ 0\times2+1\times2+3\times0 \\ 1\times0+2\times1+2\times2, \end{matrix}$ $\begin{matrix} 4+2+6=12 \\ 12\times12=24 \end{matrix}$

$O_{13} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 2 & 2 & 3 \end{bmatrix} = \begin{matrix} 2\times0+1\times1+0\times2 \\ 1\times2+3\times2+1\times0 \\ 2\times0+2\times1+3\times2, \end{matrix}$ $\begin{matrix} 1+8+8=17 \\ 17\times2=34 \end{matrix}$

Ro

$O_{21} = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 2 \\ 2 & 0 & 0 \end{bmatrix} = \begin{matrix} 2\times2+8+0=10 \\ 10\times2=20 \end{matrix}$

$O_{22} = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{matrix} 7+6+4=17 \\ 17\times2=34 \end{matrix}$

Ro

Ro. $O_{23} \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 3 \\ 0 & 2 & 2 \end{bmatrix} = \begin{matrix} 5+8+6=19 \\ 19\times2=38 \end{matrix}$

$O_{31} \begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} = \begin{matrix} 6+4+0=9 \\ 9\times2=18 \end{matrix}$

$O_{32} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{matrix} 6+0+0=6 \\ 6\times2=12 \end{matrix}$

$O_{33} = \begin{bmatrix} 2 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{matrix} 8\times4+2=14 \\ 14\times2=28 \end{matrix}$

$\therefore CO = \begin{bmatrix} 24 & 24 & 34 \\ 20 & 34 & 38 \\ 18 & 12 & 28 \end{bmatrix}$